

GATE Computer Science Core Technical Engineering Compendium

Target Exam: GATE 2026 | Stream: Engineering | Subject: Computer Science

Verified Official Academic Preparation Material - TheCampusNova Learning Portal

Key Topics Covered: Data Structures, Algorithms, OS, DBMS & Computer Networks

1. Algorithms: Asymptotic Notation (O, Omega, Theta), Master Theorem for divide-and-conquer recurrences: $T(n) = a \cdot T(n/b) + f(n)$
2. Data Structures: AVL tree rotation cases (LL, RR, LR, RL), Binary Heap insert/extract-min $O(\log n)$, Graph traversals (BFS/DFS)
3. Operating Systems: CPU Scheduling (SJF, Round Robin), Deadlock Banker Algorithm (Need = Max - Allocation $\leq$ Available)
4. DBMS: Normalization (1NF, 2NF, 3NF, BCNF lossless-join & dependency-preserving checks), Serializability precedence graph testing
5. Computer Networks: Subnetting (CIDR prefix calculations), TCP flow control vs congestion control (Slow Start, Congestion Avoidance)
6. Exam Strategy: Master Engineering Mathematics and General Aptitude first (28 marks total) to guarantee qualifying with top rank.